

COASTAL WATER QUALITY ANALYSIS OF THE BASQUE COUNTRY (1995 TO 2023)

I. INTRODUCTION

Coastal and estuarine waters face severe pressure from local pollution and climate change. This project aims to find the chemical and spatial conditions that lead to hypoxic events, periods of very low dissolved oxygen that harm marine life. Using unsupervised and supervised machine learning, the study maps water quality regimes, extracts environmental triggers, and sets predictive boundaries to observe coastal sustainability.

II. DATASET and PREPROCESSING

The data comes from the Basque Country Monitoring Network, collected between 1995 and 2023. Joining water chemistry with biological surveys (macroinvertebrates and fish) proved difficult because biological sampling is rare and station names differ by depth and tide. A strict merge gave zero matching records.

To keep the project on track, the analysis focused only on water chemistry and station locations using the WATER.csv and METADATA.csv tables. A new target variable called oxygen_status was created. A reading was labelled "Low (Stressed)" if the dissolved oxygen value fell into the lowest 25 percent of all readings (below 7.10 mg/L). The final dataset contains 5,998 rows with geographic coordinates, covering 166 monitoring stations. The main features are Temperature, Salinity, Nitrate, and Ammonium. Temperature controls how much oxygen water can hold. Salinity separates estuarine mixing zones from the open ocean. Nitrate and Ammonium indicate runoff from farming and industry.

III. EXPLORATORY DATA ANALYSIS

The EDA phase was designed to validate domain logic, check for multicollinearity, and made sure the target was balanced enough for prediction.

a. Correlation Matrix

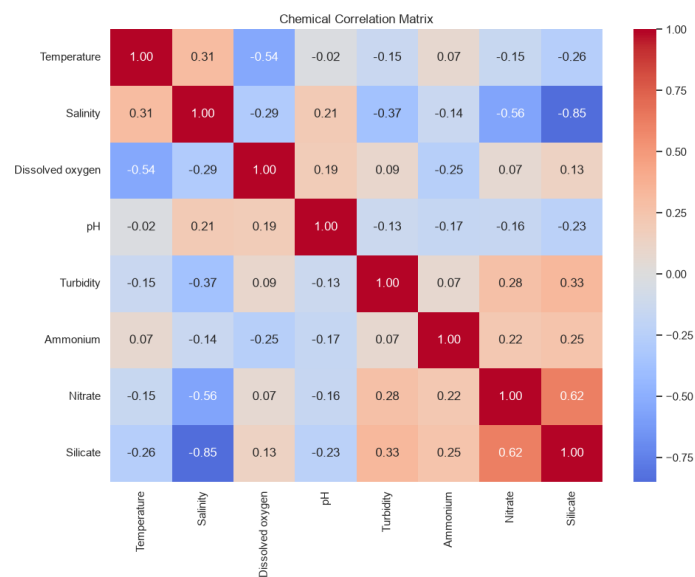


Figure 1: Chemical Correlation Matrix

The first plot was a correlation matrix. The reason for this plot was to find which chemical variables were strongly related to each other. If two variables are highly correlated, they carry similar information and one can be removed to avoid confusion in later models. The matrix showed a strong negative link (-0.54) between Temperature and Dissolved oxygen. This confirms that heat is a main cause of low oxygen. Silicate had a very strong negative link (-0.85) with Salinity. Because both describe the same effect (fresh water from rivers), Silicate was removed to keep the analysis clean.

b. Spatial Distribution

A scatter plot of station latitudes and longitudes was created to see if the sampling locations were spread out or grouped. The reason was to decide if a spatial network analysis would be useful. The plot showed that stations are not evenly spread but form natural clusters that match bays and estuaries. This confirmed that a graph analysis linking nearby stations would be effective.

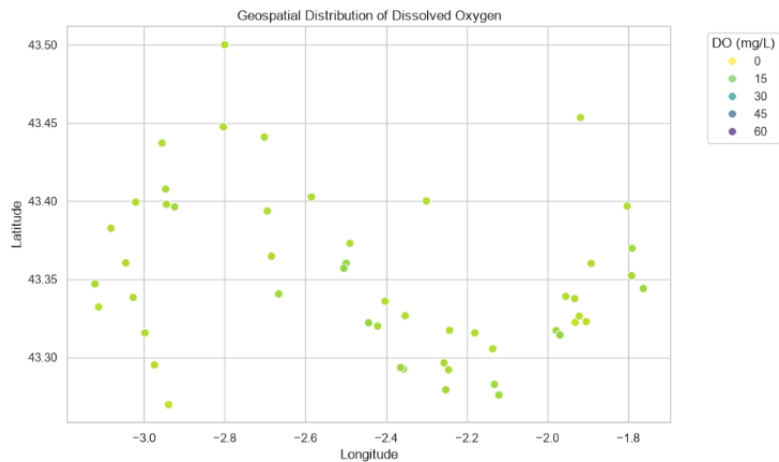


Figure 2: Geospatial Distribution of Dissolved Oxygen

c. Balance of Target Variables

A bar chart of oxygen status (Normal vs Low) was made to check how many examples of each class existed. The reason was to ensure the decision tree would have enough cases of stress to learn from. An extremely imbalanced dataset could make the tree guess only the majority class. Setting the hypoxia threshold at 7.10 mg/L gave a 75/25 split between Normal and Stressed events. This balance helped the decision tree learn from enough examples of the minority class.

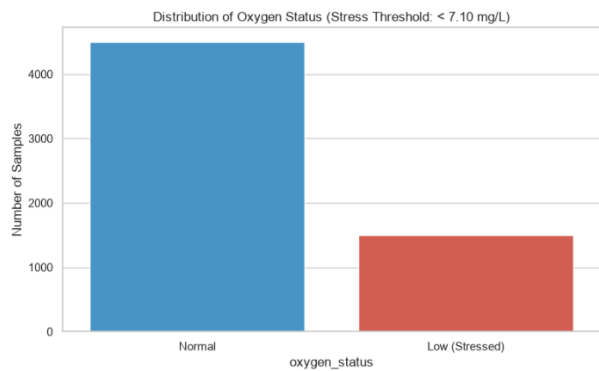


Figure 3: Distribution of Oxygen Status

d. Scatter Plots

Scatter boxplots showed how each feature was distributed for normal and stressed oxygen status. The boxes show the middle 50 percent of readings, with the median as a line inside. The scattered points show the actual spread of measurements. Stressed events had a clear upward shift in median Temperature, from about 14.5°C to 19°C. Median nutrient levels (Nitrate and Ammonium) stayed near zero for both groups, but stressed events had longer tails of high outliers. This suggests that occasional severe pollution spikes can coincide with low oxygen.

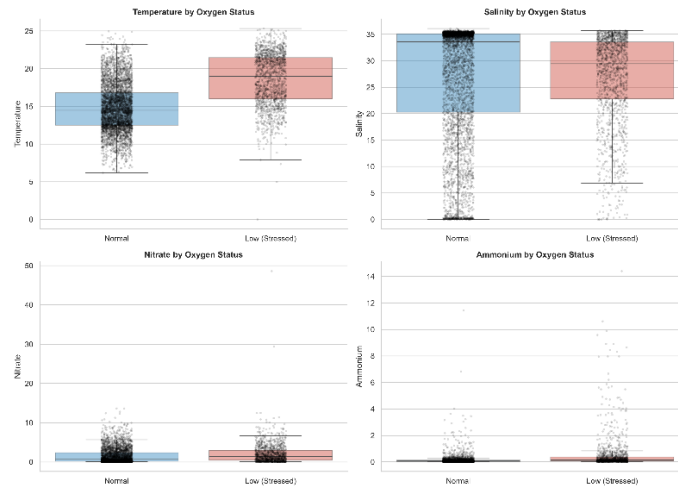


Figure 4: Feature Scatter Boxplots by Oxygen Status

IV. ANALYSIS and EVALUATION

a. K-Means Clustering

This unsupervised method aimed to see if coastal waters naturally split into hidden water quality regimes based only on their chemistry. Dual scatter plots were used. One plot compared Temperature against Salinity. The other compared Nitrate against Ammonium. The purpose of these plots was to show how the clusters separate along key chemical axes. Regime 1 formed a dense, high salinity group typical of open ocean water. Regime 0 spread across warmer, lower salinity areas typical of estuaries. The nutrient plot isolated Regime 2 as a small but extreme group with very high Ammonium and Nitrate spikes. This visual separation proves that the coast is not one uniform body. The estuarine zones hold most of the ecological stress.

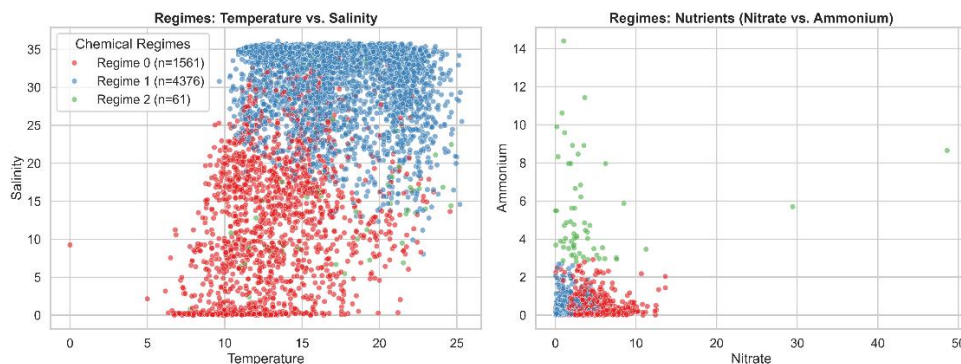


Figure 5: Dual Scatter Plots of Regimes

b. Association Rule Mining (Apriori)

The Apriori algorithm was used to find frequent combinations of environmental categories that occur together with ecological stress. This method works well with categorical data to find "baskets" of conditions that appear together more often than by chance. The strength of each rule was measured by Lift, which shows how many times more likely the outcome is when the conditions are present, compared to the outcome's normal chance. After turning continuous numbers into categories, the algorithm produced 63 rules. The strongest rule had a Lift of 3.54 for the combination of Medium Salinity and High Temperature. This proves that coastal hypoxia is not random. An ecosystem is about 3.5 times more likely to suffer severe stress when summer heatwaves meet estuarine mixing zones.

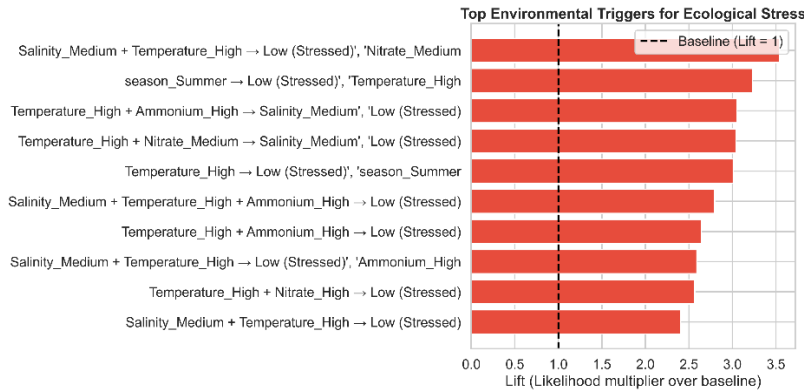


Figure 6: Top Environmental Triggers for Ecological Stress

c. Decision Tree Classification

The decision tree was used to find the exact number values that trigger hypoxia, creating a clear rule for environmental management. The tree made its first split at exactly 16.27°C. A secondary plot confirmed that most low oxygen readings lie to the right of this temperature line.

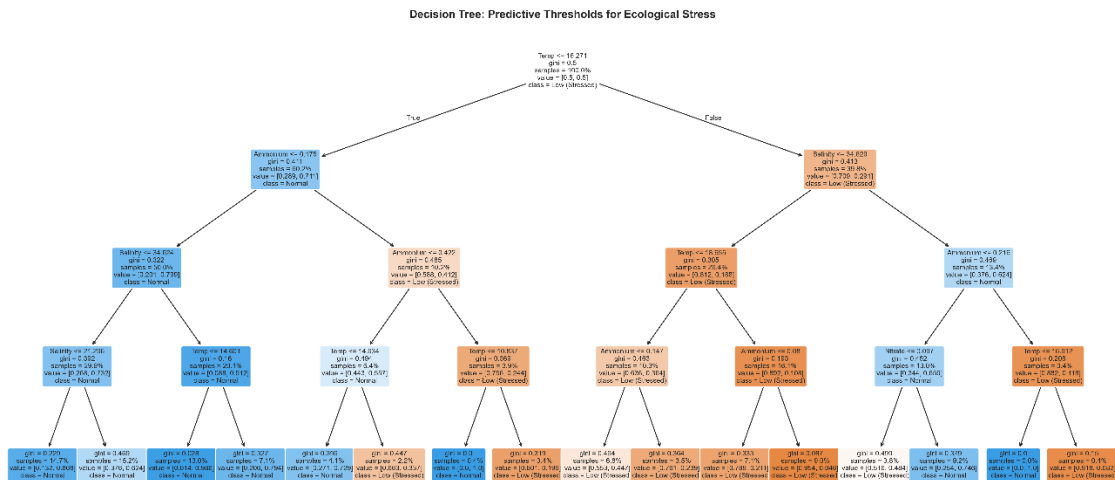


Figure 7: Decision Tree Structure

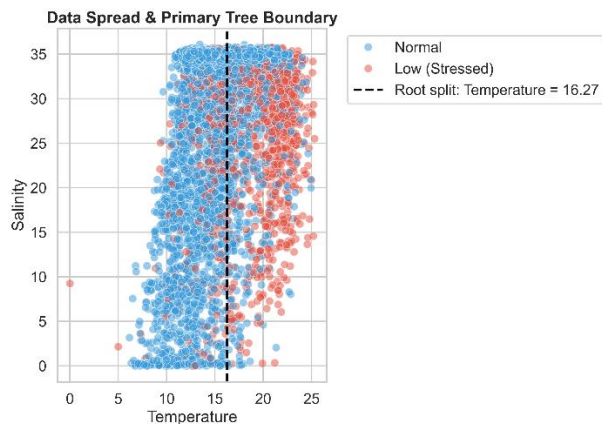


Table 1: Classification Report

CLASS	PRECISION	RECALL	F1- SCORE	SUPPORT
LOW (STRESSED)	0.54	0.79	0.64	300
NORMAL	0.92	0.78	0.84	900
ACCURACY			0.78	1200

Figure 8: Data Spread & Primary Tree Boundary

The model had an overall accuracy of 78 percent. The most important number is the recall of 0.79 for the stressed class. In tracking environmental conditions, catching dangerous events is more important than avoiding false alarms. The precision of 0.54 shows the model is somewhat aggressive, but it still catches 79 percent of all hypoxic events. This makes 16.27°C a useful early warning trigger for monitoring systems.

d. Graph Analysis

Graph analysis was used to test whether poor water quality is an isolated local problem or a connected regional issue. Station coordinates were treated as nodes, and each station was linked to its nearest neighbors. This created a network of 51 nodes and 153 edges. The nodes were colored according to the K-Means regimes found earlier. The resulting visual shows that hypoxia spreads among neighboring stations. Stressed estuarine zones form tight groups inside sheltered bays. This means pollution management must target specific vulnerable hubs rather than apply broad regional rules.

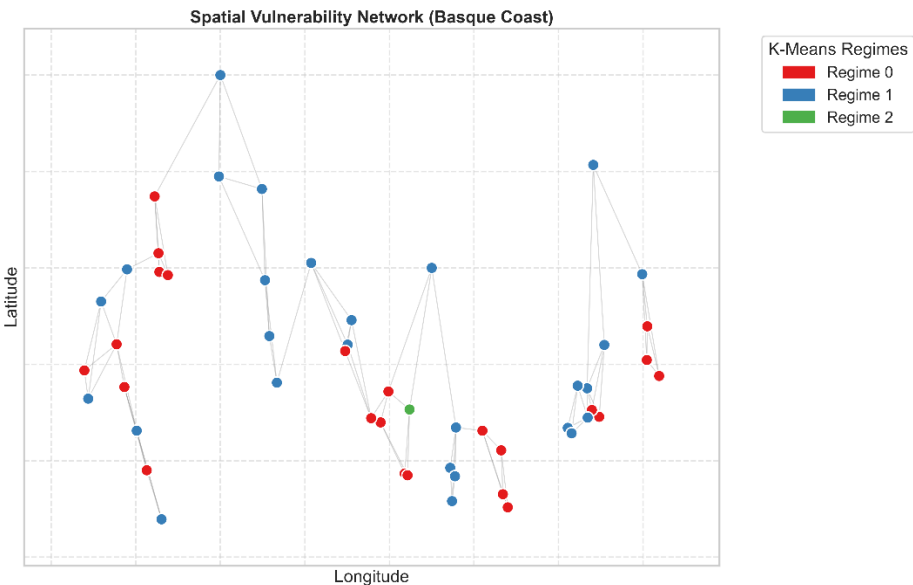


Figure 9: Spatial Vulnerability Network

e. Temporal Trend Analysis

To see if the ecosystem is stable or worsening, the frequency of "Low (Stressed)" events was tracked across the full 30 years of data. The scatter line plot shows a volatile but clearly rising trend in ecological stress. In the mid 1990s, the area averaged 30 to 40 hypoxic events per year. The trend line shows that the baseline has shifted upward, and recent years often record 60 to 80 severe stress events. This increase means that local ecosystems are crossing the previously found thermal boundaries more often. This directly reflects the growing local impact of global climate change.

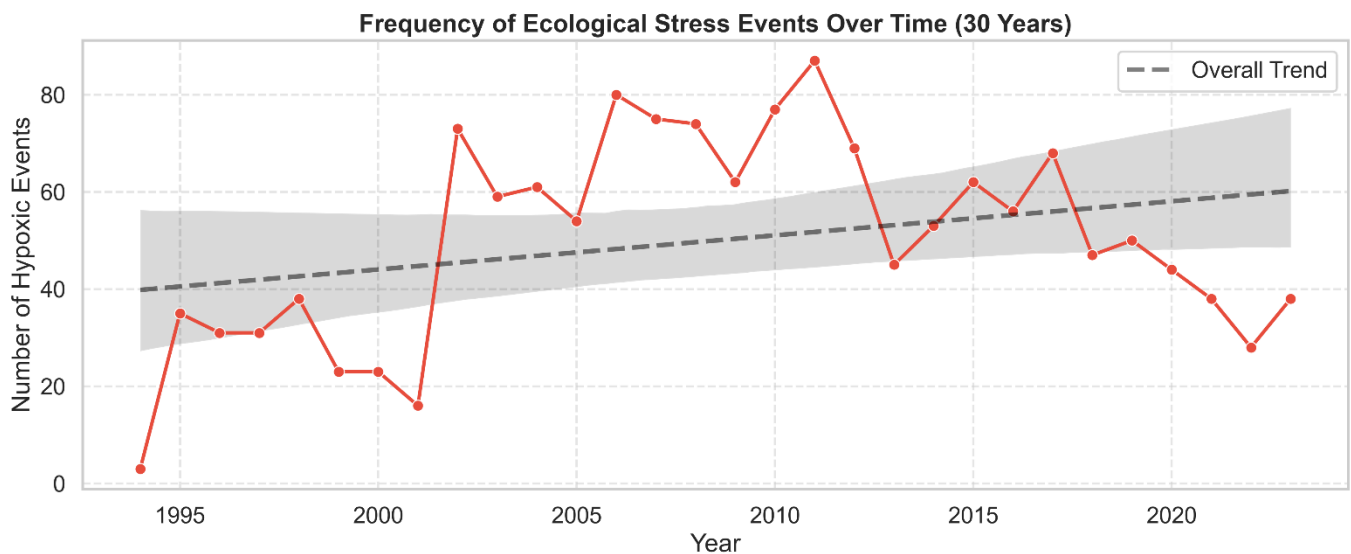


Figure 10: Frequency of Ecological Stress Events Over Time

V. CONCLUSION

This project combined spatial coordinates with water chemistry to build a full risk profile of the Basque Country coast. By linking unsupervised clustering, association rule mining, and a decision tree, the analysis showed that estuarine waters crossing a strict temperature threshold of 16.27°C have the highest risk of hypoxic collapse. The spatial network also confirmed that these risks are not scattered randomly but form local, connected danger zones inside sheltered bays. These data driven insights give environmental agencies a clear, practical framework for setting up targeted early warning monitoring systems in a changing climate.

REFERENCES

Borja Á, Adarraga I, Bald J, Belzunce-Segarra MJ, Cruz I, Franco J, Garmendia JM, Larreta J, Laza-Martínez A, Manzanos A, Marquiegui MA, Martín I, Martínez J, Menchaca I, Pouso S, Revilla M, Rodríguez JG, Ruiz JM, Sagarmínaga Y, Solaun O, Uriarte A, Zorita I and Muxika I (2024). Marine Biodiversity and Environmental Data: An AI-Ready, Open Dataset from the long term (1995–2023) Basque Country Monitoring Network. *Front. Ocean Sustain.* 2:1528837. doi: 10.3389/focsu.2024.1528837

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